Projection

it is a **dimensionality reduction technique** that transforms high-dimensional data into a lower-dimensional space while preserving the essential information

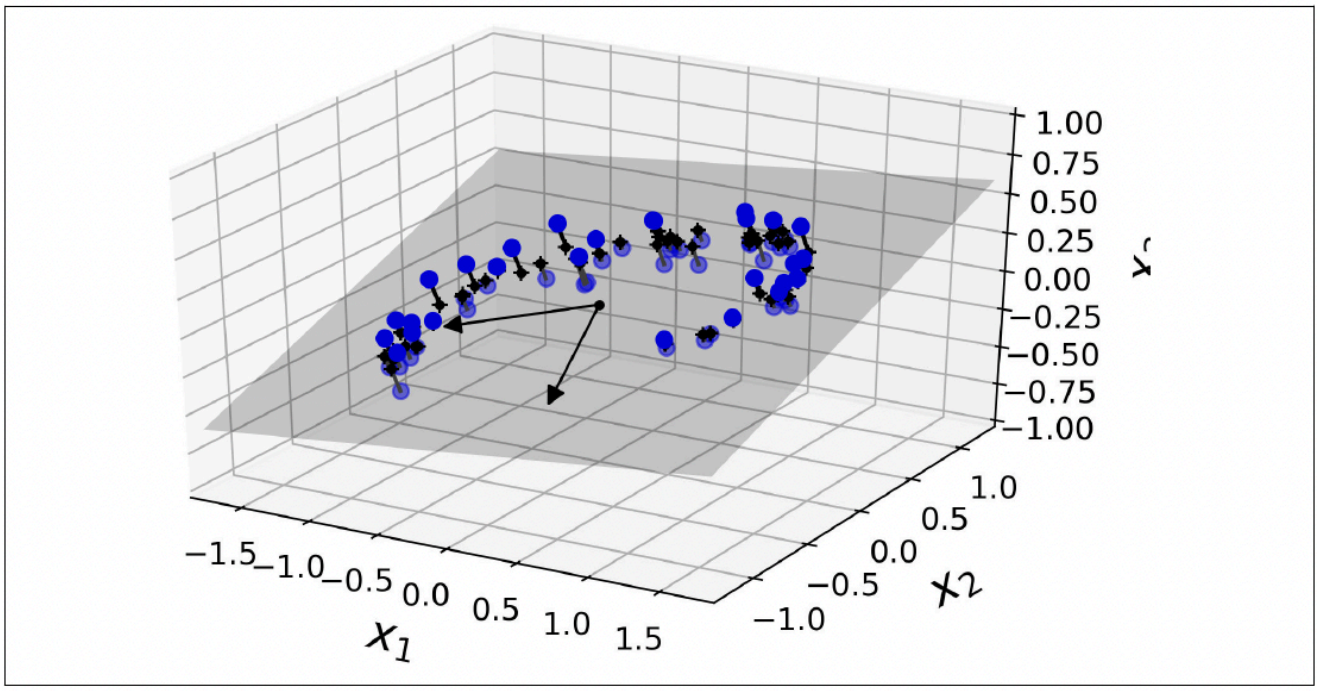

Figure 8-2. A 3D dataset lying close to a 2D subspace

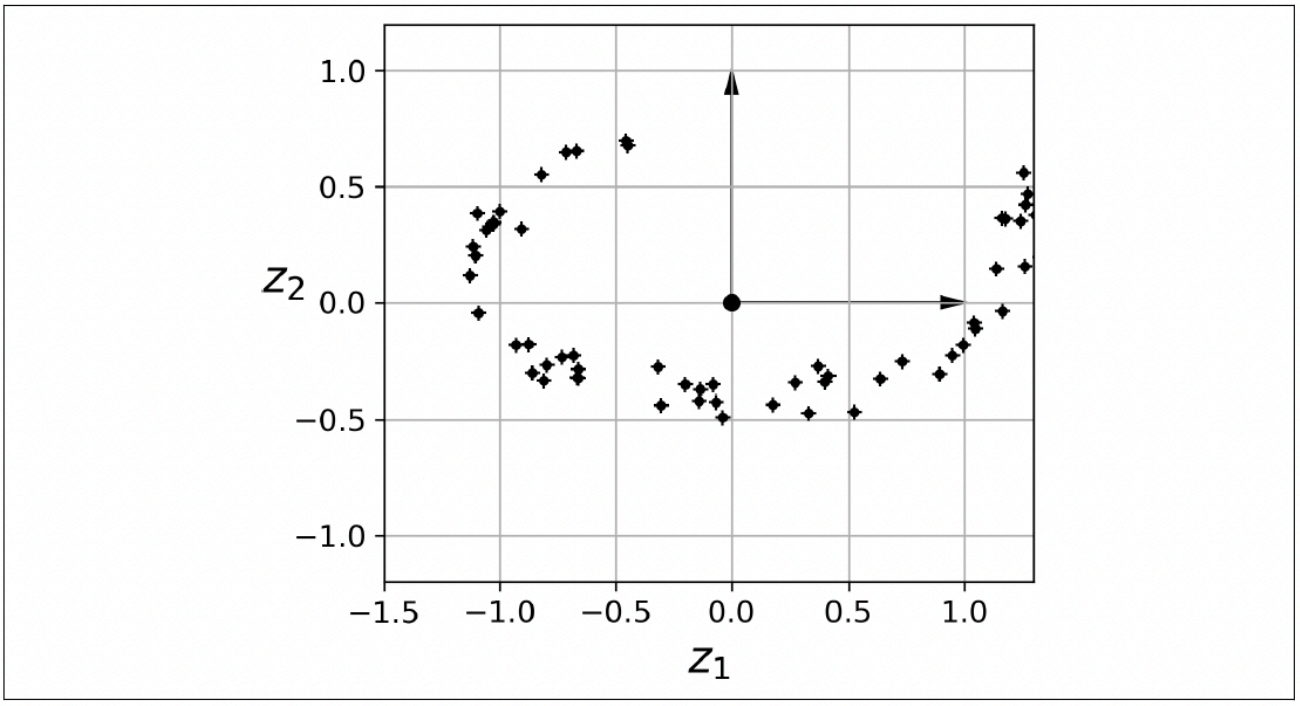

Figure 8-3. The new 2D dataset after projection

But, it is not good for the complex higher dimensional space

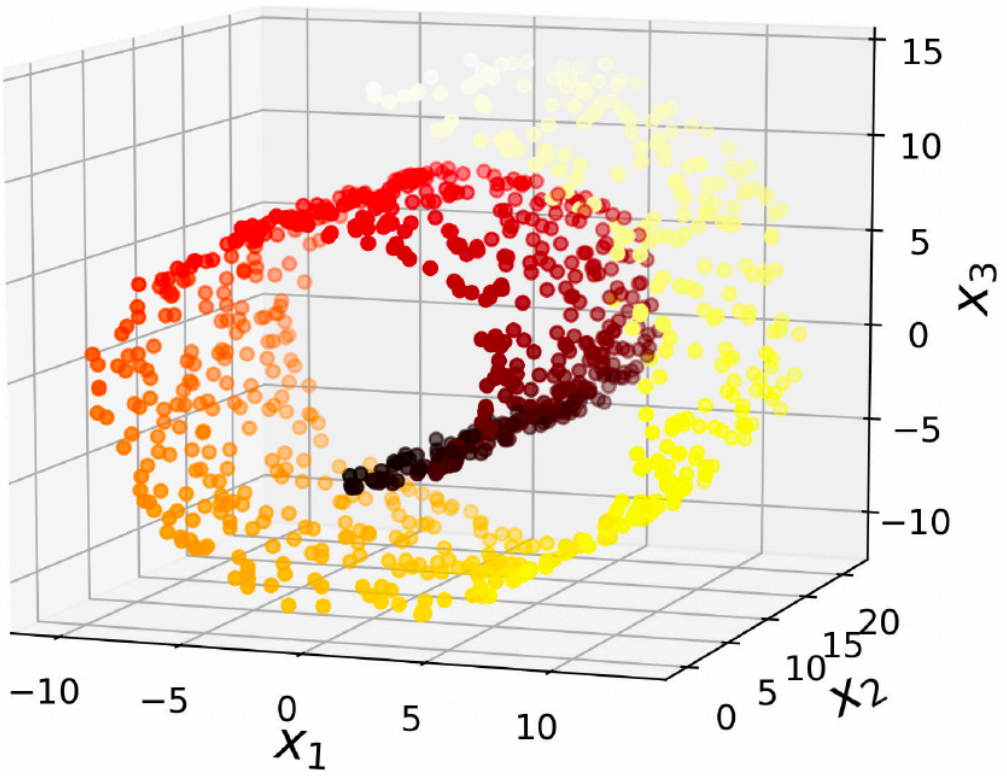

Figure 8-4. Swiss roll dataset

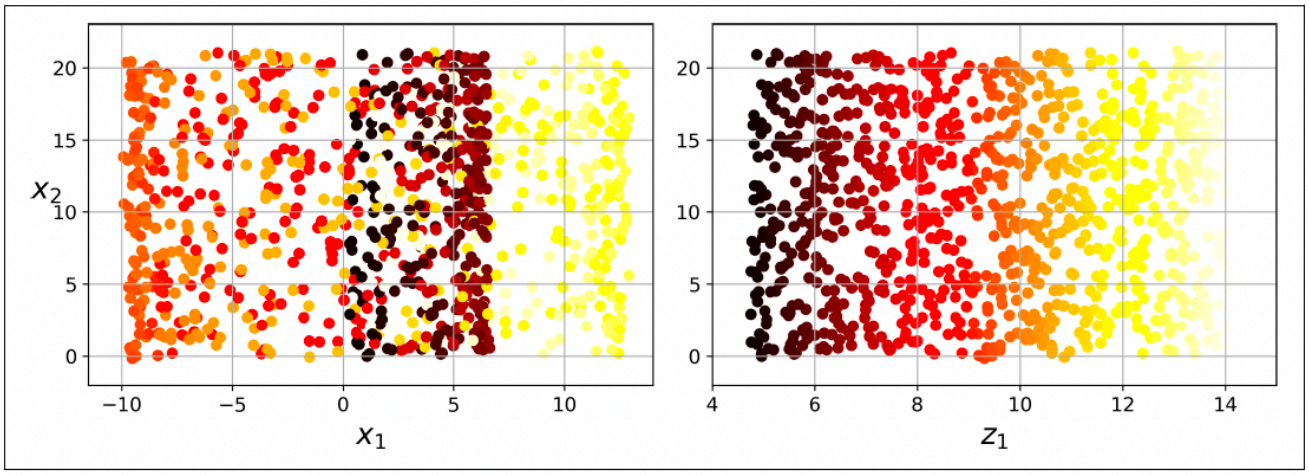

Figure 8-5. Squashing by projecting onto a plane (left) versus unrolling the Swiss roll (right)